Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-based real-time quality control and qualification in large-format additive manufacturing**			
Korea's 12 National Strategic Technology Fields	AI (√), Advanced Robotics & Manufacturing (√)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Machinery and materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Machine Automation Technology	Robotics/ Automation Machinery	Other Robot/Automation Machinery Technology
Concept of Proposed Technology	The proposed technology aims to develop an AI-based real-time quality control system for large-format additive manufacturing (AM) that provides continuous process awareness, predictive quality assessment, and corrective action during deposition process. **1. Main functions** • AI-based in-situ process monitoring using multimodal sensor data • AI-based predictive model for early prediction of defect occurrence and quality deviations • Adaptive process control for real-time adjustment of control variables to stabilize the process			

	and improve part precision **2. Potential uses** • Real-time quality control and process stabilization for AM systems • Manufacturing high-reliability, large-scale AM components for sectors such as aerospace, energy, defense, maritime • Reduction of trial-and-error tuning and improved repeatability for customized parts or low-volume production
Rationale for Joint R&D Project	Large-format additive manufacturing (LFAM) is a cornerstone technology for producing customized, large-scale components in high-value industries. However, broader industrial adoption is constrained by high process variability, frequent defect occurrence, and the lack of reliable process control capabilities. Accordingly, AI-based quality control system enabling continuous monitoring and in-process correction is critical to achieving stable, repeatable, production-ready large-format AM. Such a system cannot be developed and deployed without closely coordinated expertise and infrastructure, necessitating collaboration between a research institute and an industrial partner with complementary strengths: • **Research institute (Georgia Tech):** Provides advanced capabilities in AI modeling, multimodal high dimensional data analytics, multistage process and quality modeling, and the development of predictive and adaptive control methodologies. In particular, Professor Jianjun Shi (Endowed Chair Professor at School of ISyE at Georgia Tech, a member of National Academy of Engineering of USA, https://sites.gatech.edu/jianjun-shi/) is globally recognized as a pioneer in In-Process Quality Improvement (IPQI) and Stream of Variation (SoV) methodologies. His research integrates system informatics and control theory with data science to optimize complex manufacturing systems by fusing engineering models with advanced analytics [1]. With funding from major government agencies (NSF, NIST, Department of Energy) and global industry leaders (GM, Ford, Boeing, Lockheed-Martin, Samsung, Apple), his group has successfully translated research into technologies deployed across diverse production systems with substantial economic impact. Of particular relevance to this project is the deep expertise in (i) extracting actionable process signatures from heterogeneous multi-sensor data collected from complex manufacturing system such as additive manufacturing [2,3,10], (ii) building physics-informed and statistical–AI hybrid models that connect evolving process states to point-cloud–level geometric deviation, defect formation, and functional qualification [4-6], (iii) designing uncertainty-aware prediction and closed-loop control in learned embedding spaces to regulate high-dimensional, partially observed manufacturing processes [7-9]. • **Korean SME partner:** Provides access to AM platforms and drives the integration of in-situ sensing hardware with controller-level interfaces for real-time actuation. By facilitating systematic on-machine testing within operating environments, the partner enables the acquisition of high-quality, representative datasets - including off-nominal cases - needed to develop, calibrate, and validate the proposed technologies.

	Expected benefits of joint R&D • Direct integration of advanced algorithmic research with industrial hardware bridges the gap between laboratory prototyping and factory-floor deployment, ensuring developed techniques and control logic are immediately applicable to real-world AM operations. • Rigorous validation in production environments ensures the system meets strict precision, reliability, and repeatability requirements demanded by high-stakes industry sectors. • This collaboration elevates the Korean SME partner's global competitiveness, establishing a credible foundation to penetrate high-value international markets and secure overseas business opportunities.
Key Words	1. Multimodal in-situ monitoring 2. Physics-informed AI-based predictive modeling 3. Adaptive process control 4. Real-time quality control 5. Large-format additive manufacturing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	**The primary objective** of this project is to develop an AI-based real-time quality control system for large-format AM, integrating in-situ process monitoring, predictive defect modeling, and adaptive process control. **The specific R&D objectives are:** 1. To develop in-situ process monitoring based on multimodal learning algorithm to provide multi-stage process states 2. To develop physics-informed learning models for forecasting potential defects, quality deviations, and process instabilities, enabling proactive intervention 3. To develop adaptive process quality control algorithm for real-time adjustment of key control variables			
R&D Target Specifications	**The target specifications** for the R&D Objectives are as follows:			

Objectives	Performance parameter	Target specification
Multimodal in-situ monitoring	Anomaly detection accuracy	90%
Physics-informed predictive quality modeling	Defect & instability classification performance	90%
Adaptive process control	Quality outcome improvement (reduction in defect-related events)	≥ 50% (more than 20% reduction in defect-related events when printing the same component.)

R&D Plan	The project will develop an AI–based real-time quality control system for large-format AM through a staged integration of (1) multimodal in-situ monitoring, (2) physics-informed predictive quality modeling, and (3) adaptive process control. The development follows an iterative build–test–refine cycle, progressing from algorithm development and laboratory validation to on-machine deployment and industrial-scale validation at the Korean SME partner site. Year 1: Multimodal in-situ process monitoring 1. Key tasks & methodologies • Develop robust pipelines for synchronization, calibration, and preprocessing of high-frequency multimodal sensor streams (IR thermal, optical vision, and 3D spatial profiles) • Establish denoising and feature extraction protocols to handle high-dimensional data streams without compromising system latency • Implement multimodal learning algorithm to estimate in-process quality states, including a high-fidelity representation of the evolving quality and real-time detection of anomalous process behavior 2. Expected milestones • Synchronized multimodal data acquisition and processing pipeline • Multimodal learning algorithm validated for accurate in-process quality state estimation and anomaly detection Year 2: Physics-informed predictive modeling and adaptive process control 1. Key tasks & methodologies • Develop physics-informed learning model that integrates thermos-mechanical principles to forecast potential defects, quality deviations, and process instabilities • Develop adaptive process control algorithm capable of adjusting key control variables (feed rate, travel speed, heat input, and others) in real time based on monitored process states and predicted quality states 2. Expected milestones • Physics-informed model validated for predicting future defects and quality deviations • Demonstration of the adaptive control on the SME platform Year 3: System deployment, validation, and technology advancement 1. Key tasks & methodologies • Integrate monitoring, prediction, and control modules into a unified quality control system suitable for deployment • Conduct comprehensive experiments across representative parts and operating conditions to evaluate performance, repeatability, robustness, and reliability, and refine the system based on results • Prepare technology transfer documentation to support industrial adoption 2. Expected milestones • Developed quality control system verified to operate reliably within the Korean SME's production environment • Industrial validation report and transfer package completed, including benchmark results and deployment guidelines

Research and Development Timeline

Year & Objective	Tasks	Quarter											
		Q1			Q2			Q3			Q4		
Year 1: Multimodal in-situ process monitoring	Multimodal data acquisition												
	Processing pipeline												
	Multimodal algorithm												
Year 2: Physics-informed predictive modeling and adaptive process control	Physics-informed predictive modeling												
	Adaptive process control												
Year 3: System deployment, validation, and technology advancement	Unified deployable architecture												
	Validation and improvement												
	Technology transfer package												

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

The team at Georgia Tech is responsible for the development of core AI algorithms, theoretical frameworks, and validation methodologies required for the proposed quality control technology in large-format AM process.

1. Technical contributions
   - Core AI algorithms for multimodal process monitoring, physics-informed predictive modeling and adaptive process control
   - An integrated quality control system combining developed algorithms
2. Research scope
   - Define the overall research architecture and system framework for the AI-based quality control
   - Design experimental protocols, benchmark test cases, and performance metrics for algorithm training and evaluation
   - Perform analytical studies to support maturation of developed technologies prior to and during on-machine deployment
3. Resources to be provided
   - Research personnel with expertise in data science, AI/ML, manufacturing systems, and quality engineering
   - Access to advanced AI frameworks and specialized analytics tools for high-dimensional data processing
4. Specific tasks to be provided

	• Technical guidance and close collaboration with the Korean SME partner to support developed system integration, on-machine experimentation, and performance evaluation
Expected R&R of Korean SME Partner	**The Korean SME partner** is responsible for providing the large-format AM platform, sensing and control integration environment, and validation infrastructure required to implement and demonstrate the proposed AI-based real-time quality control system 1. Technical contributions • Installation and synchronization of in-situ sensing hardware within the AM platform to enable multimodal data collection • Establishment of controller-level interfaces required for real-time actuation and process adjustments • Deployment of the developed quality control onto the large-format AM platform 2. Research scope • Define problem statements, representative parts, settings, defect modes, and benchmark test cases • Generate and provide high-quality datasets, including both nominal and off-nominal (defect) cases for model development and validation • Execute trials and long-duration fabrication experiments to evaluate system performance • Provide practical feedback and support iterative refinement of developed quality control 3. Resources to be provided • Access to industrial-grade large-format additive manufacturing platforms and integrated sensing equipment • Technical staff with expertise in AM operation, automation, and manufacturing process management • A dedicated testing environment for validation 4. Specific tasks to be provided • Facilitation of technology transfer and preparation for commercial-scale implementation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes and impact of this joint R&D project 1. Technological achievements • Delivery of a field-ready AI-based quality control system that integrates multimodal in-situ monitoring, predictive quality modeling, and real-time adaptive process control for high-precision large-format AM • Comprehensive verification of system performance through industrial-scale fabrication, demonstrating robustness in a realistic production environment 2. Commercialization potentials • The integration of autonomous quality control will provide a decisive competitive edge for the Korean start-up or SME partner's hardware, increasing equipment ROI for end-users and accelerating the adoption of AM • The developed technology can be commercialized as an integrated monitoring-and-control software suite, an add-on package, or a standalone module for existing large-format AM platforms • Establishment of high-margin business models, including subscription-based services and premium options for next-generation manufacturing systems 3. Industrial or economic impact • Developed AI-driven technology will significantly reduce scrap rates, rework, and machine downtime, leading to optimized throughput and a substantial increase in "first-pass yield" for large-scale production • Reduced scrap, rework, and downtime improve throughput and first-pass yield, lowering

	production cost for large-format AM • Enhanced process stability and repeatability will enable LFAM to meet the rigorous quality standards required for the aerospace, energy, defense, and maritime sectors, opening new markets for customized large-scale components • The scalable and transferable nature of the developed AI framework ensures its extensibility to other manufacturing domains, fostering a more resilient, high-tech, and self-sufficient domestic SME ecosystem • This collaboration provides Korean SMEs with differentiated, AI-enabled capabilities, directly supporting the national agenda for AI Transformation (AX) and facilitating entry into high-value-added global markets • The partnership with Georgia Tech, a world-leading research institute, establishes a global standard for autonomous manufacturing. This strengthens Korea's international technological standing and develops a pipeline of advanced manufacturing talent

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization